

Construction of DFA- Part I

Complete Course On Theory Of Computation For GATE CS & IT

CS & IT Engineering



**Lecture Number
06**



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Topics to be Covered

1

2

3

Minimal DFA for InFinite Language

Construct the minimal DFA that Accept all the string of 0's and 1's

i) including ϵ

ii) excluding ϵ

Soln: 1. $L = \Sigma^* = \{ \epsilon, 0, 1, 00, 01, 10, 11, 000, \dots \}$

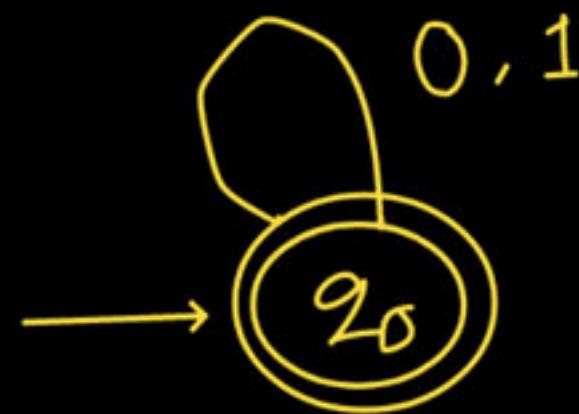
Note

DFA accept infinite
Language GF and

only GF gt contains 2.

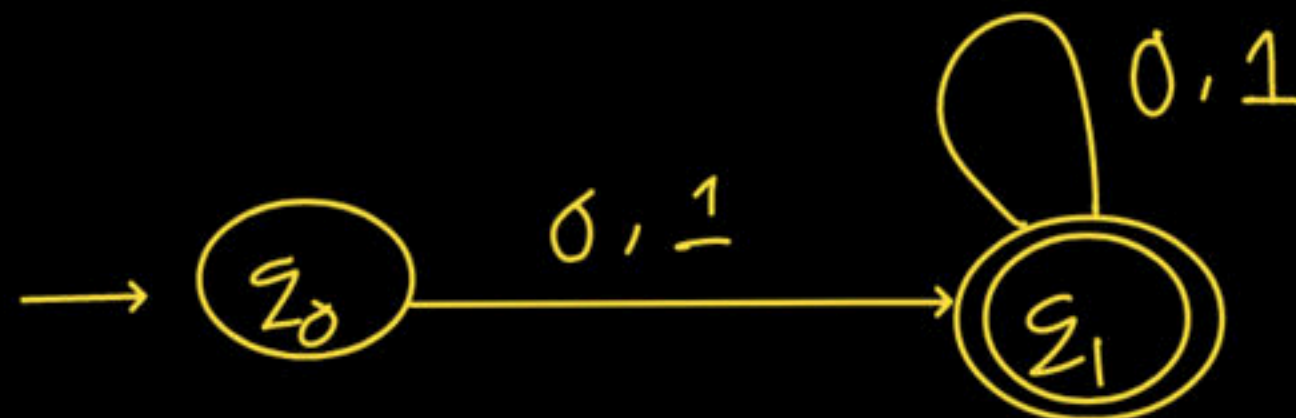
Self Loop or cycle

{ Graph with productive
state }



states in mDFA = 1

2. $L = \Sigma^+ = \{ 0, 1, 00, 01, 10, 11, \dots \}$



states in mDFA = 2

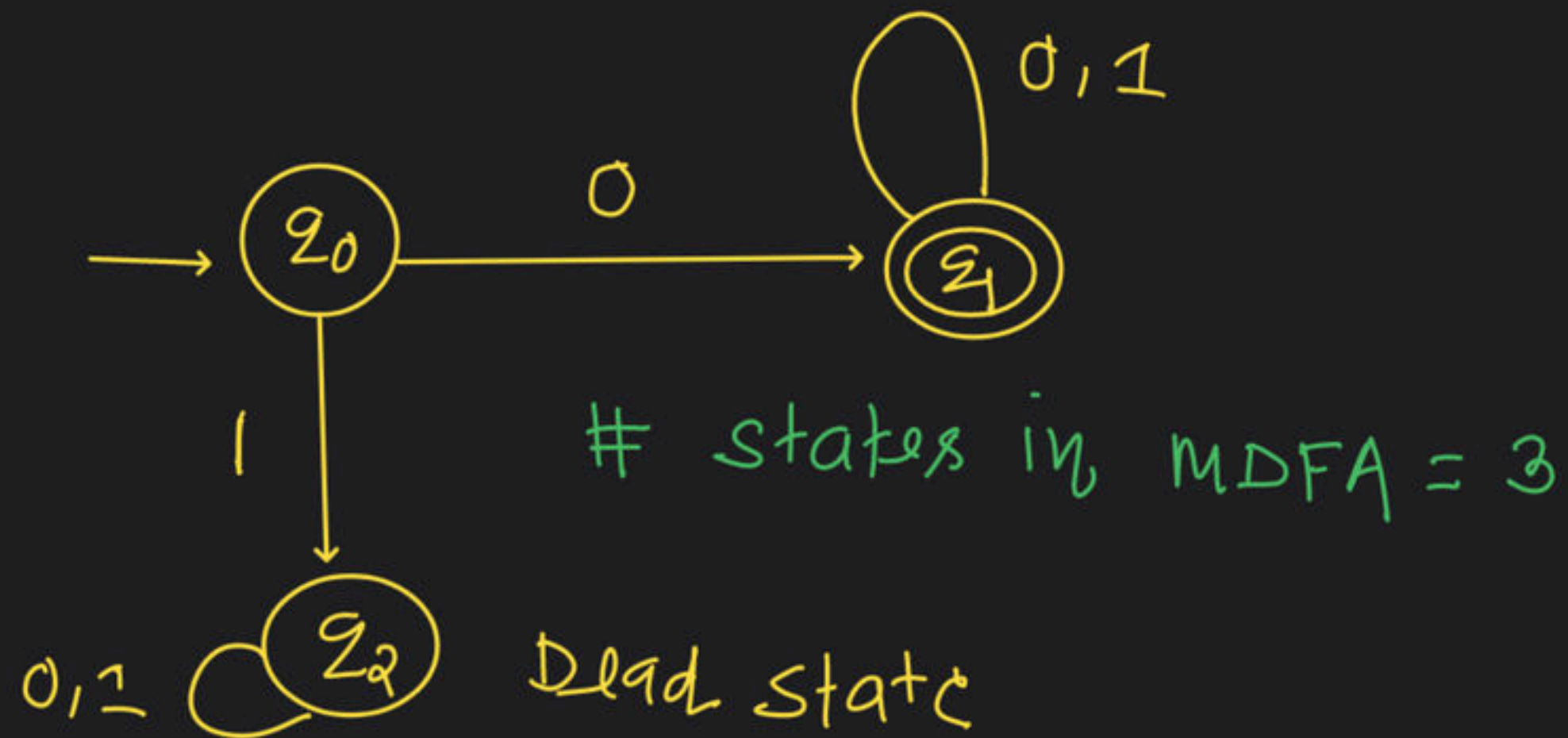
Construct the minimal DFA that Accept all the string of 0's and 1's where every string

- i) start with 0 $\longrightarrow$ 3 state $\longrightarrow$ $1 + 2 = 3$ state
- ii) start with 10 $\longrightarrow$ 4 state $\longrightarrow$ $2 + 2 = 4$ state
- iii) start with 010 $\longrightarrow$ 5 state $\longrightarrow$ $3 + 2 = 5$ state
- iv) start with 1010 $\longrightarrow$ 6 state $\longrightarrow$ $4 + 2 = 6$ state
- v) start with 00110 $\longrightarrow$ 7 state $\longrightarrow$ $5 + 2 = 7$ state

(i) Soln: Every string start with 0

$$L = 0X, \Sigma = \{0, 1\}$$

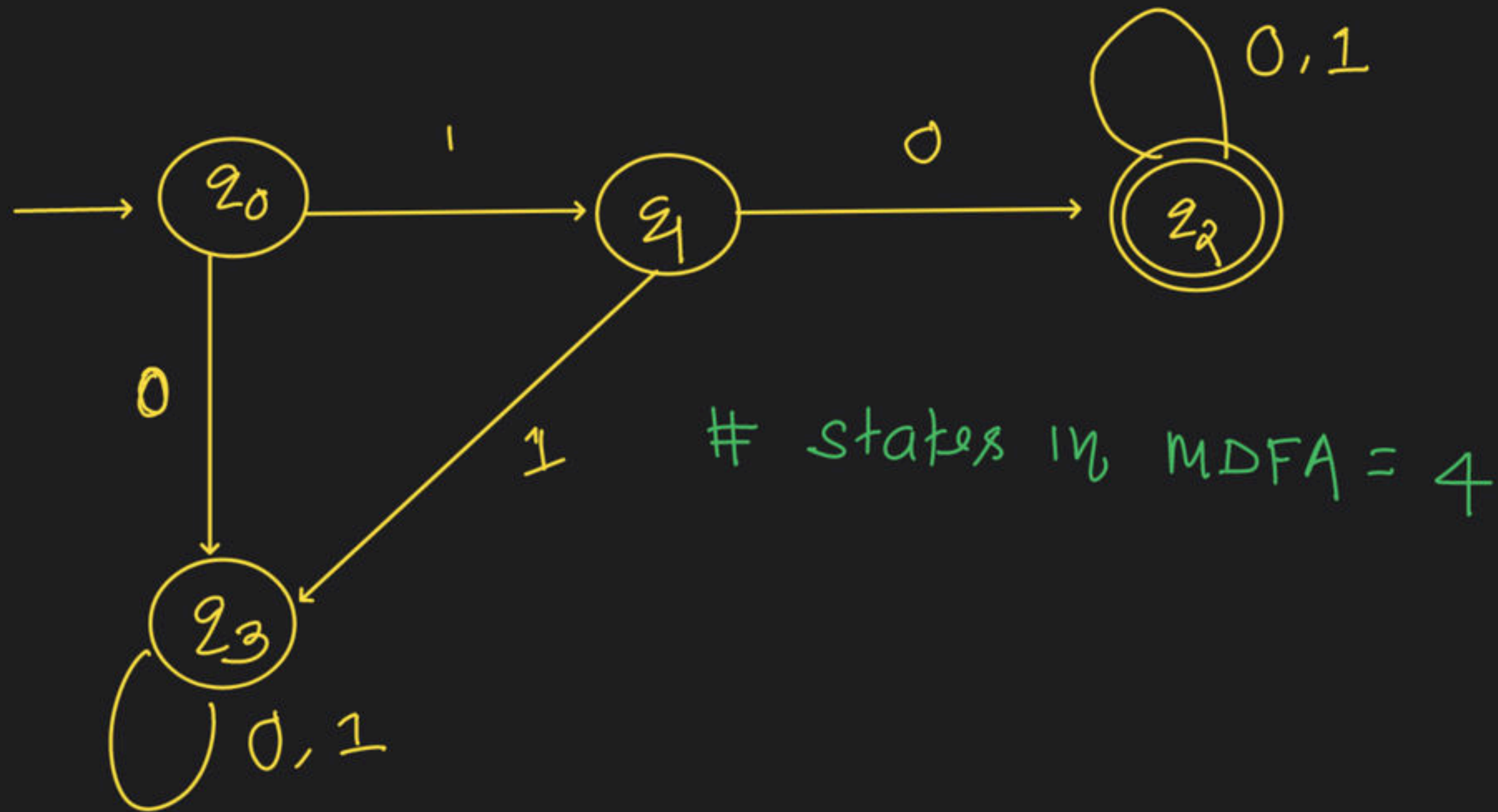
$$L = 0, 00, 01, 000, 001, 010, 011, \dots$$



(ii) Soln: Every string start with 10

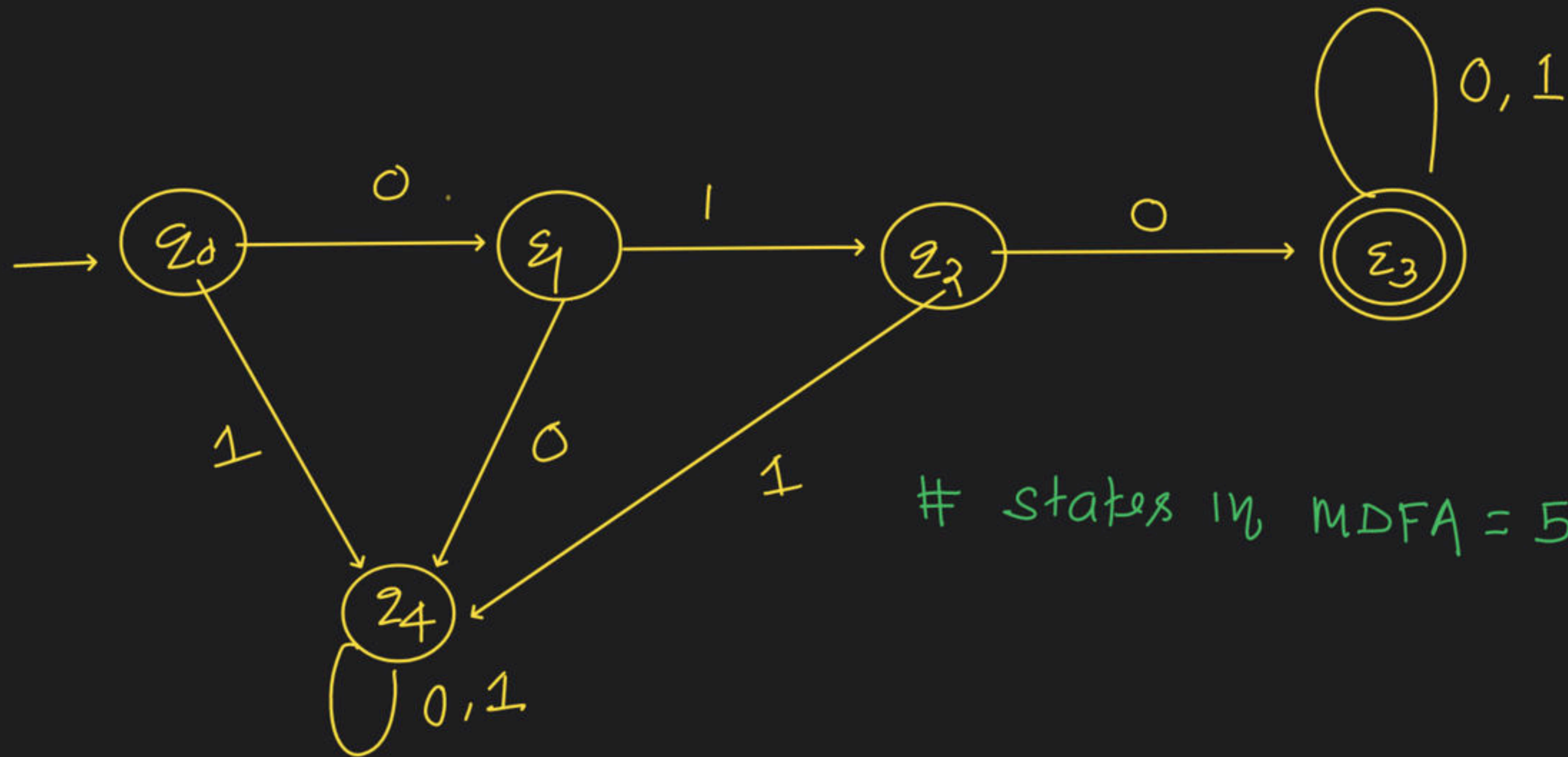
$$L = 10X, \Sigma = \{0, 1\}$$

$$L = 10, 100, 101, \dots$$



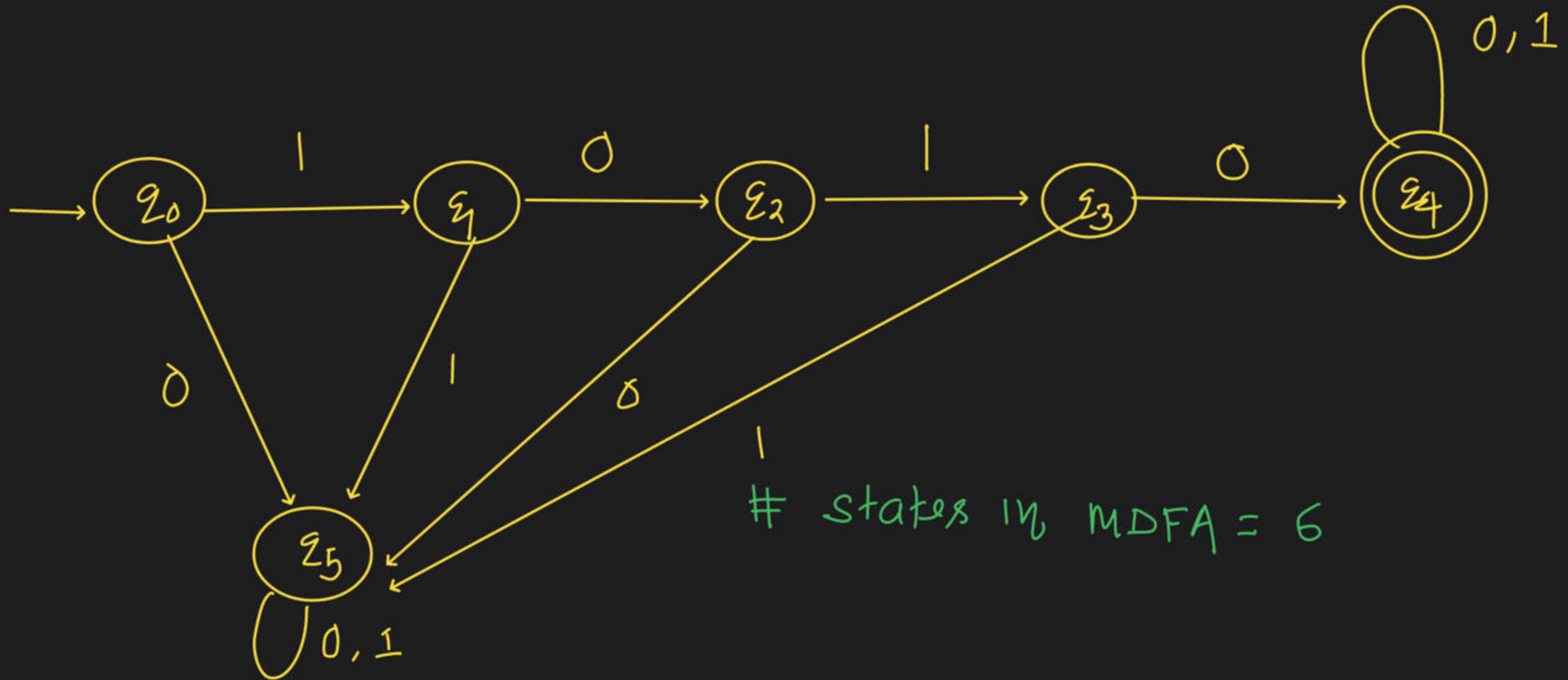
(iii) Soln: Every string start with 010

$$L = 010X, \Sigma = \{0, 1\}$$



(iv) soln: Every string start with 1010

$L = 1010X$, $\Sigma = \{0, 1\}$



Let L is the set of all string of 0's and 1's where every string starts with a substring S of length n i.e $L=SX$, $|s|=n$. Then number of states in minimal DFA to accept this language is $n+2$

Construct the minimal DFA that Accept all the string of 0's and 1's where

- ✓ i) **2nd symbol from left end is 0** $\longrightarrow$ 4 state $\longrightarrow 2+2 = 4$ state
- ✓ ii) **3rd symbol from left end is 1** $\longrightarrow$ 5 state $\longrightarrow 3+2 = 5$ state
- ✓ iii) **4th symbol from left end is 0** $\longrightarrow$ 6 state $\longrightarrow 4+2 = 6$ state
- ✓ iv) **5th symbol from left end is 1** $\longrightarrow$ 7 state $\longrightarrow 5+2 = 7$ state

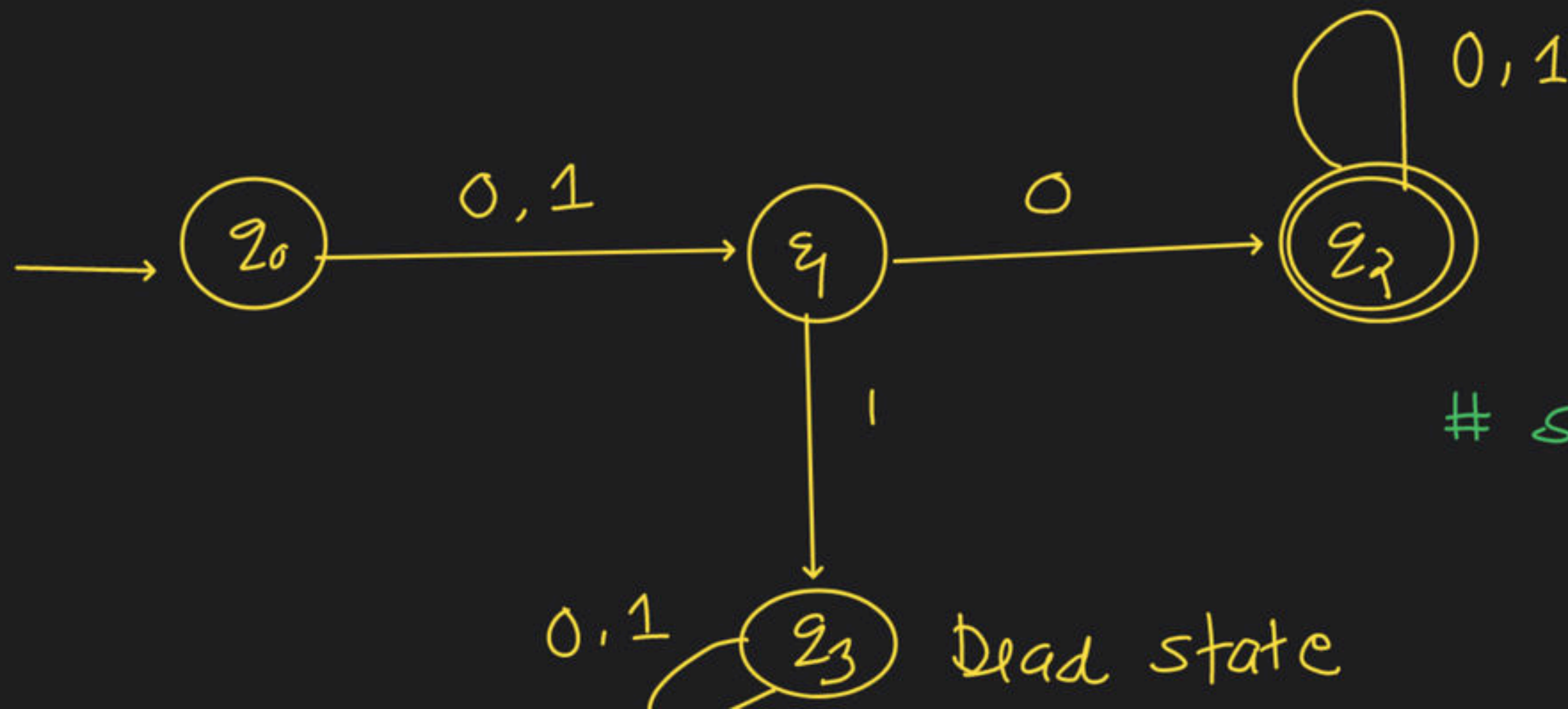
(i) Soln: and symbol from Left end is 0

$L = X \textcircled{0} X X X \text{-----}$

↓ ↓ ↓ ↓ ↓

$\{0, 1\}$

$L = \{00, 10, 001, 101\}$



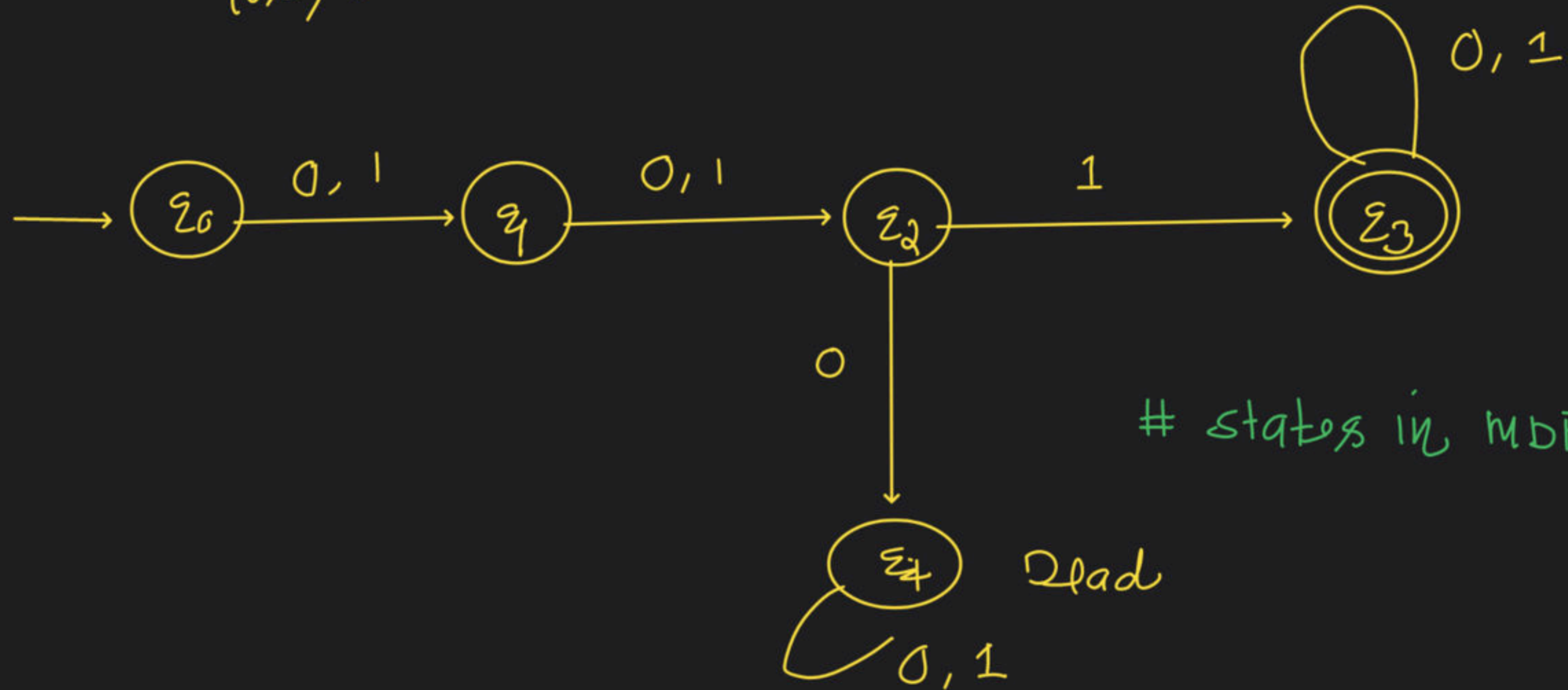
states in DFA = 4

(ii) Solⁿ: 3rd symbol From Left end is 1

$L =$ $\times$ $\times$ 1 $\times$ $\times$ $\times$ -----

$\downarrow$ $\downarrow$

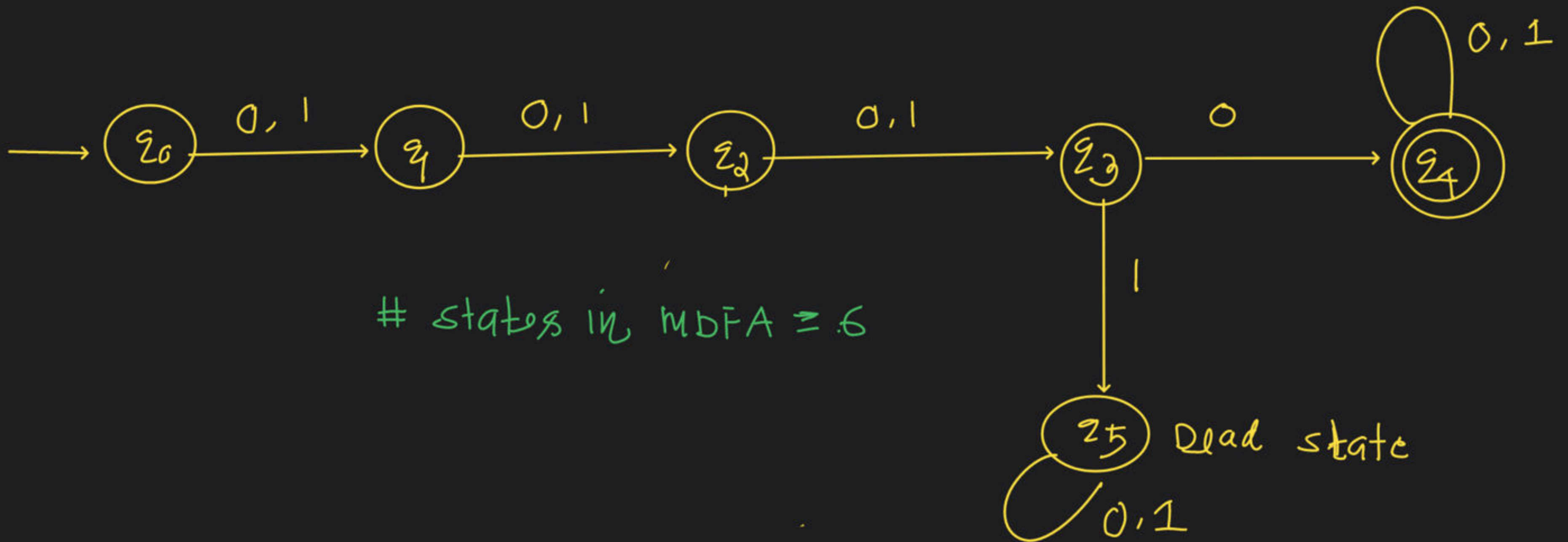
$\{0,1\}$ $\{0,1\}$



states in DFA = 5

(iii) Soln: 4th symbol From Left end is 0

$L = X \ X \ X \ 0 \ X \ X \ X \ \dots$
 ↓ ↓ ↓ ↓
 0 0 0 0



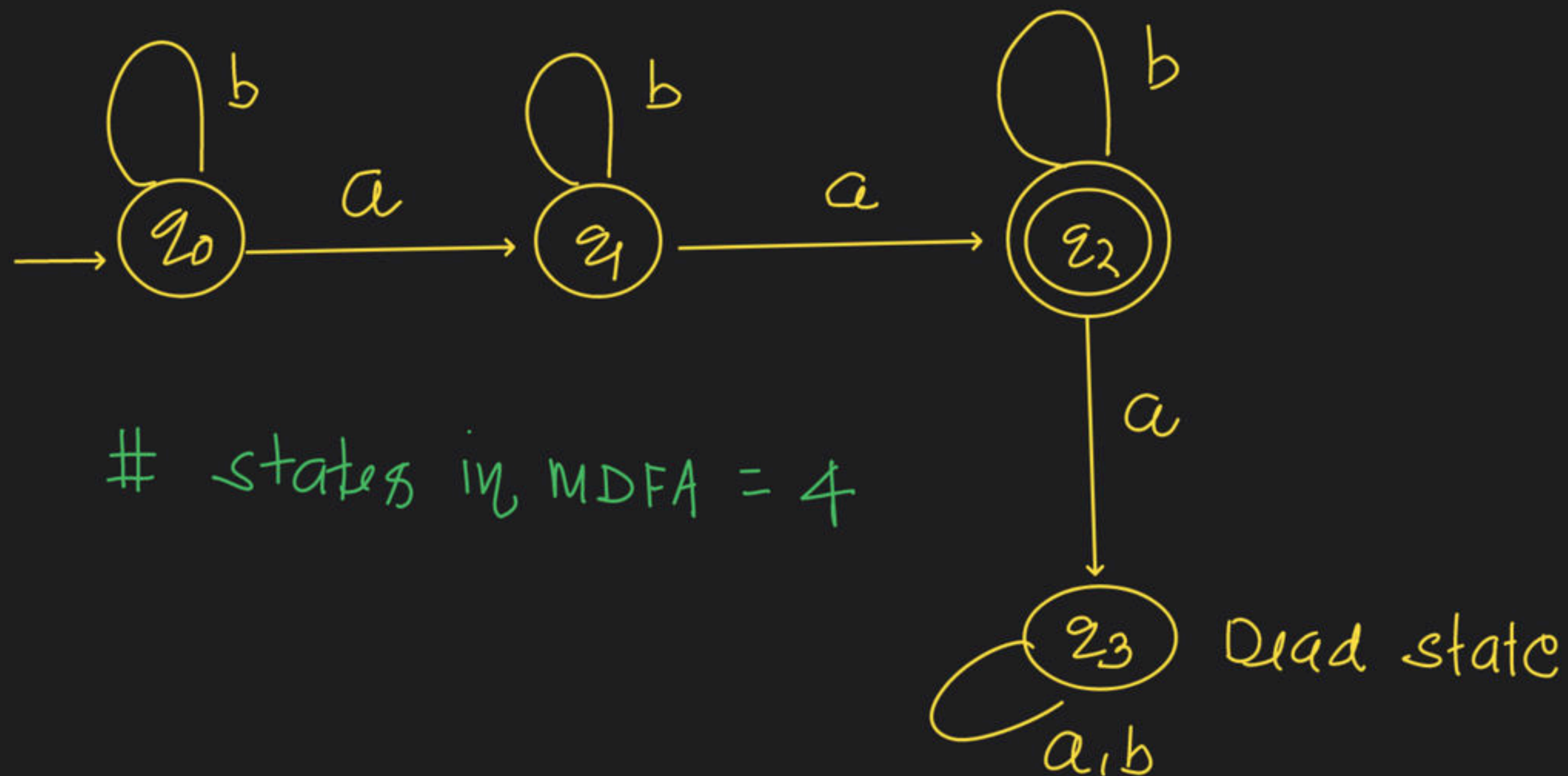
Let L is the set of all string of 0's and 1's where n th symbol from left end is fixed . Then number of states in minimal DFA to accept this language is $n+2$

Construct the minimal DFA that Accept all the string of a's and b's where

- ✓ i) No. of a's exactly 2 $\longrightarrow$ 4 state $\longrightarrow 2+2 = 4$ state
- ✓ ii) No. of a's almost 2 $\longrightarrow$ 4 state $\longrightarrow 2+2 = 4$ state
- ✓ iii) No. of a's atleast 2 $\longrightarrow$ 3 state $\longrightarrow 2+1 = 3$ state

(i)

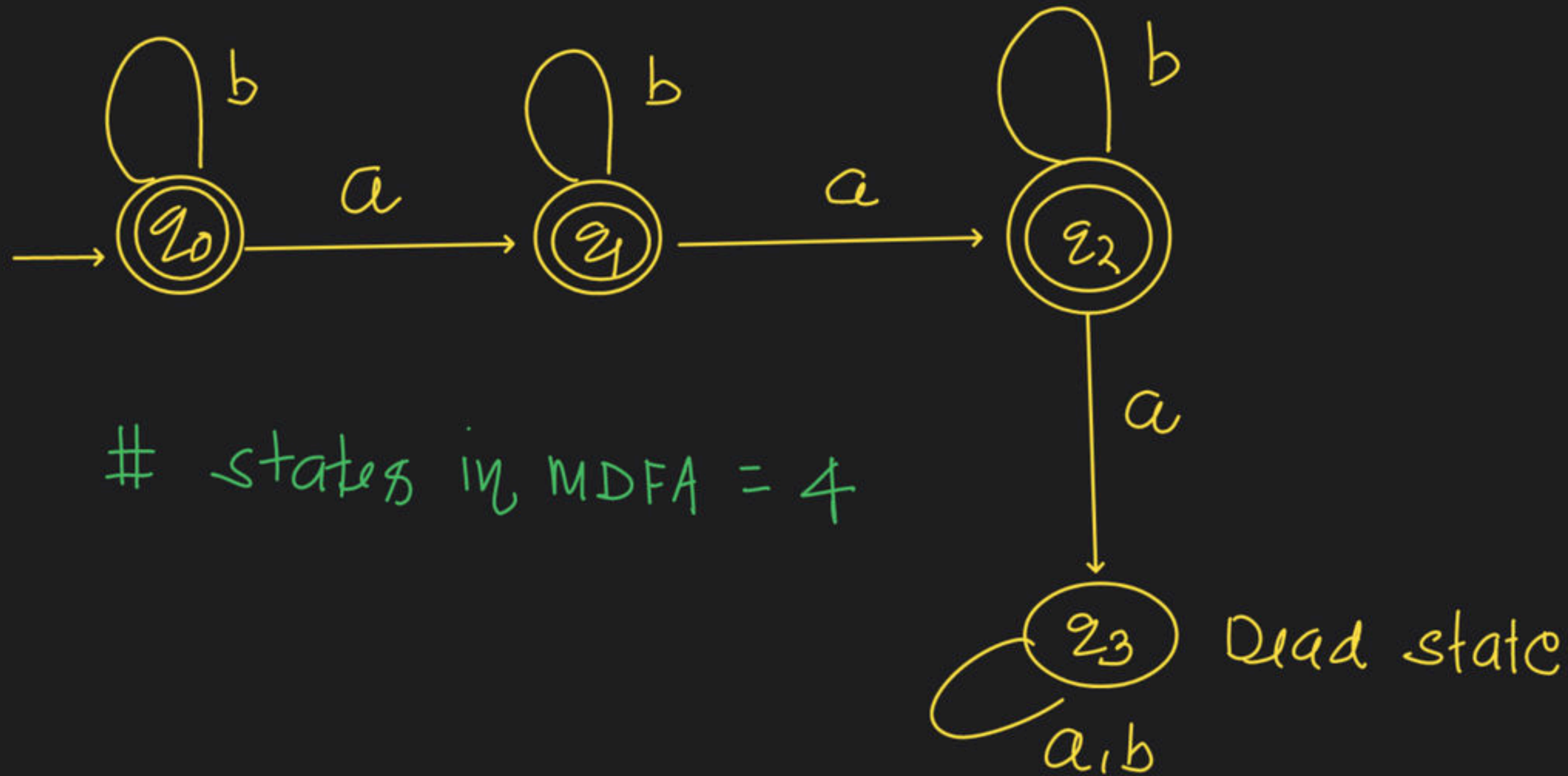
$$|w|_a = 2, \Sigma = \{a, b\}$$



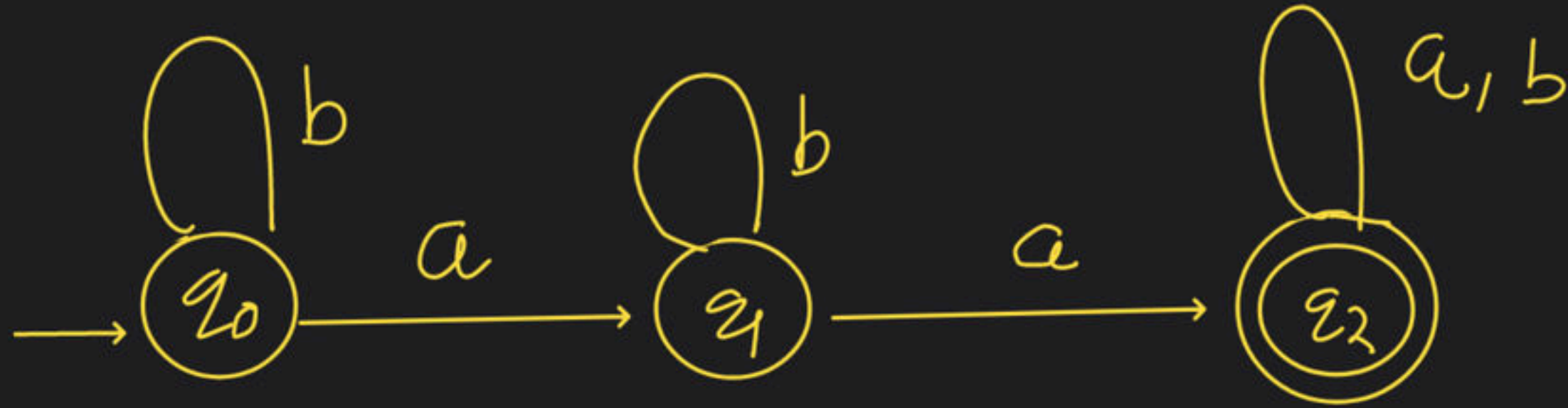
(ii)

$|w|_a \leq 2, \Sigma = \{a, b\}$

$\downarrow$
0, 1, 2



(ii) $|w|_a \geq 2$, $\Sigma = \{a, b\}$
 $\downarrow$
 $a, 3, 4$



states in MDFA = 3

Note {ADRule}

$$|w|_a = n \longrightarrow n+2 \text{ state}$$

$$|w|_a \leq n \longrightarrow n+2 \text{ state}$$

$$|w|_a \geq n \longrightarrow n+2 \text{ state}$$

Note {ADRule}

$$|w|_b = n \longrightarrow n+2 \text{ state}$$

$$|w|_b \leq n \longrightarrow n+2 \text{ state}$$

$$|w|_b \geq n \longrightarrow n+1 \text{ state}$$

Construct the minimal DFA that Accept all the string of a's and b's where

- ✓ 1) No. of a's are even
- ✓ 2) No. of b's are odd
- ✓ 3) No. of a's congruent of 2 (Mod 3)
- ✓ 4) No. of b's congruent of 3 (Mod 4)
- ✓ 5) No. of a's congruent of 2 (Mod 5)
- ✓ 6) No. of b's congruent of 3 (Mod 6)
- ✓ 7) No. of a's congruent of 2 (Mod 7)

(i) Soln:

No. of a 's even

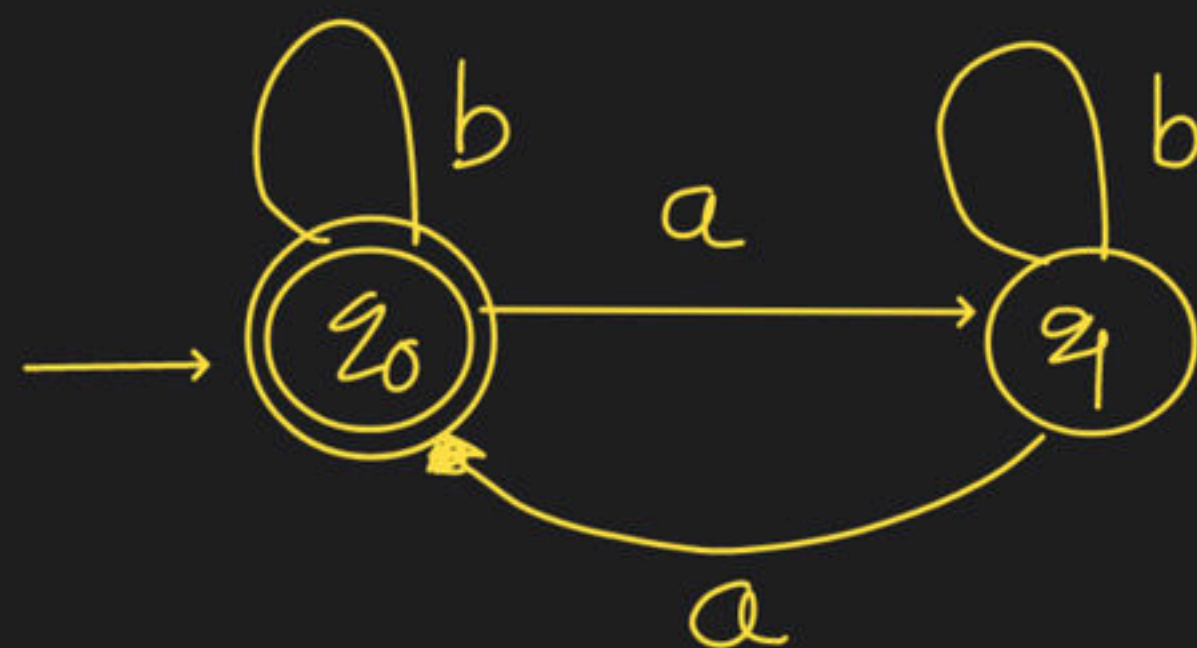
$|w|_a = \text{even}$

$|w|_a = 0 \pmod{2} = 0, 2, 4, 6, 8, 10, \dots$

Rem

0 1

q_0 q_1

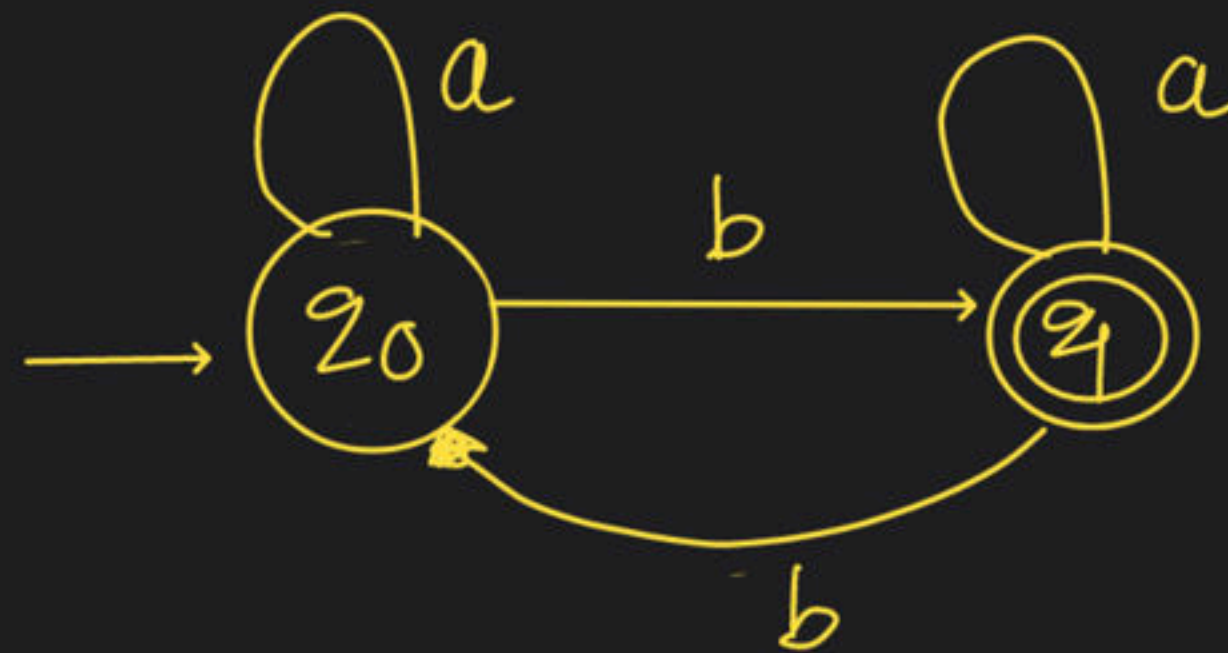
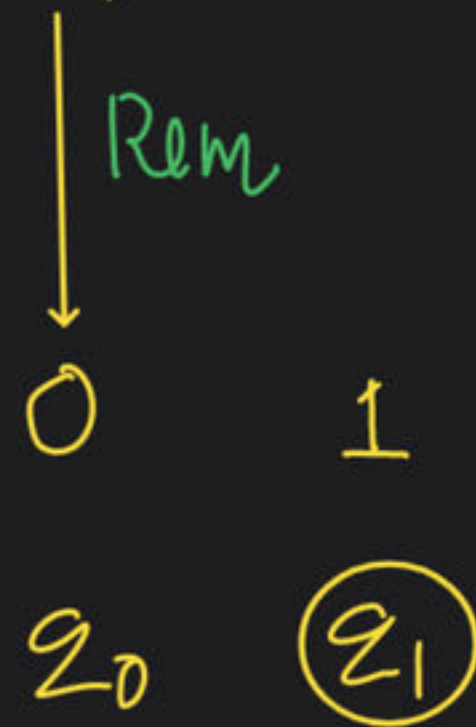


(ii) Soln:

No. of b's odd

$$|w|_b = \text{odd}$$

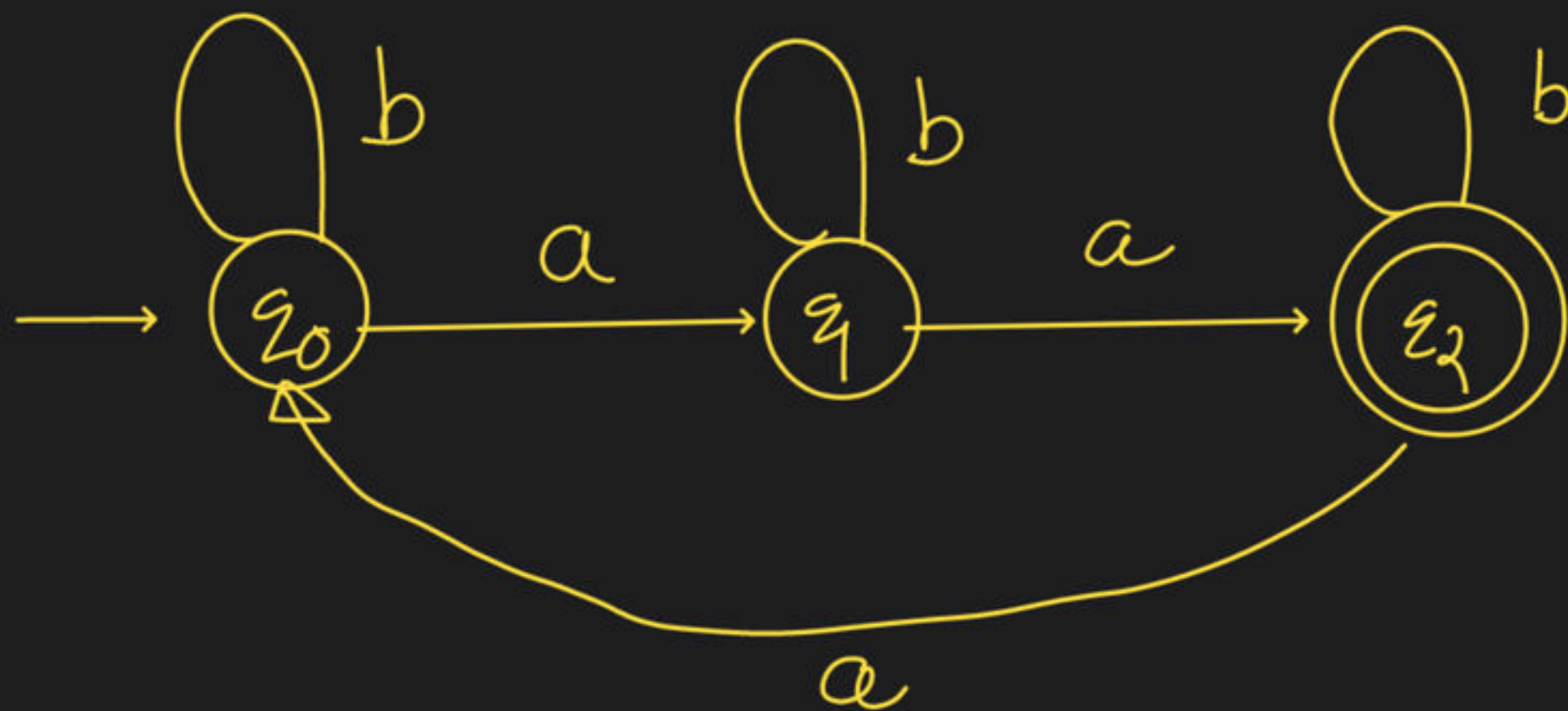
$$|w|_b = 1 \pmod{2} = 1, 3, 5, 7, 9$$



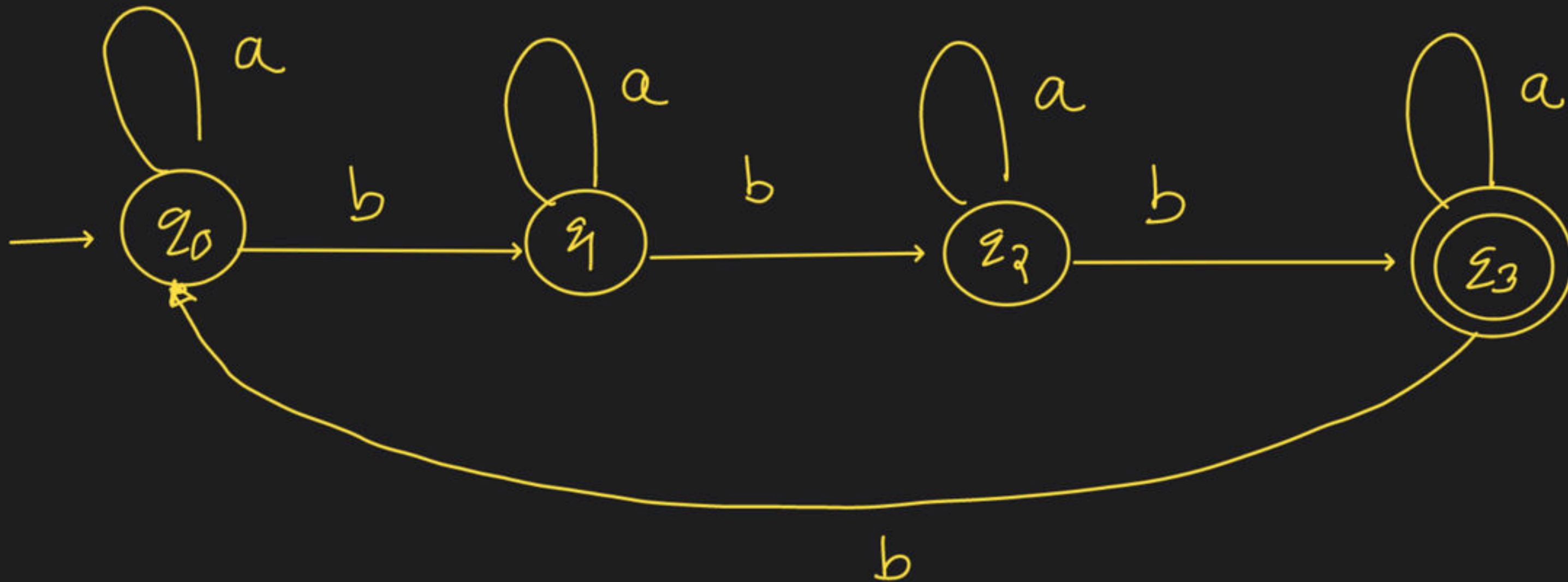
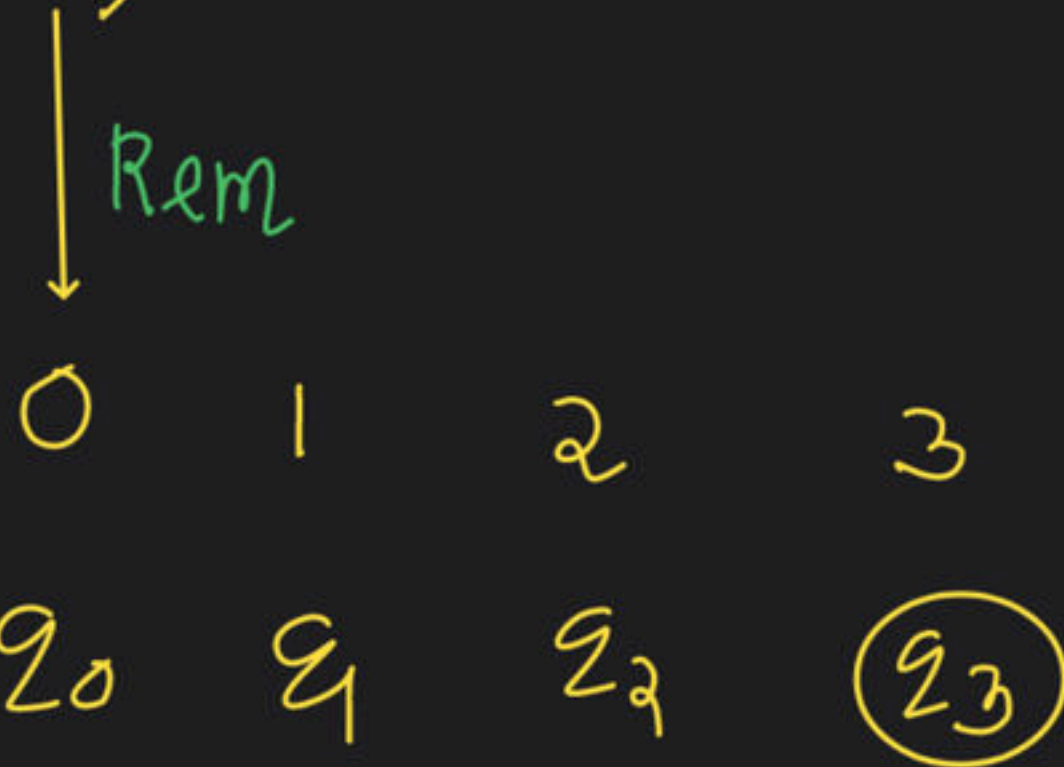
$$|w|_a \equiv 2 \pmod{3} = \underbrace{2, 5, 8, 11, 14, 17, 20}_{\substack{3 \quad 3 \quad 3}}$$

Rem
0 1 2

q_0 q_1 (q_2)



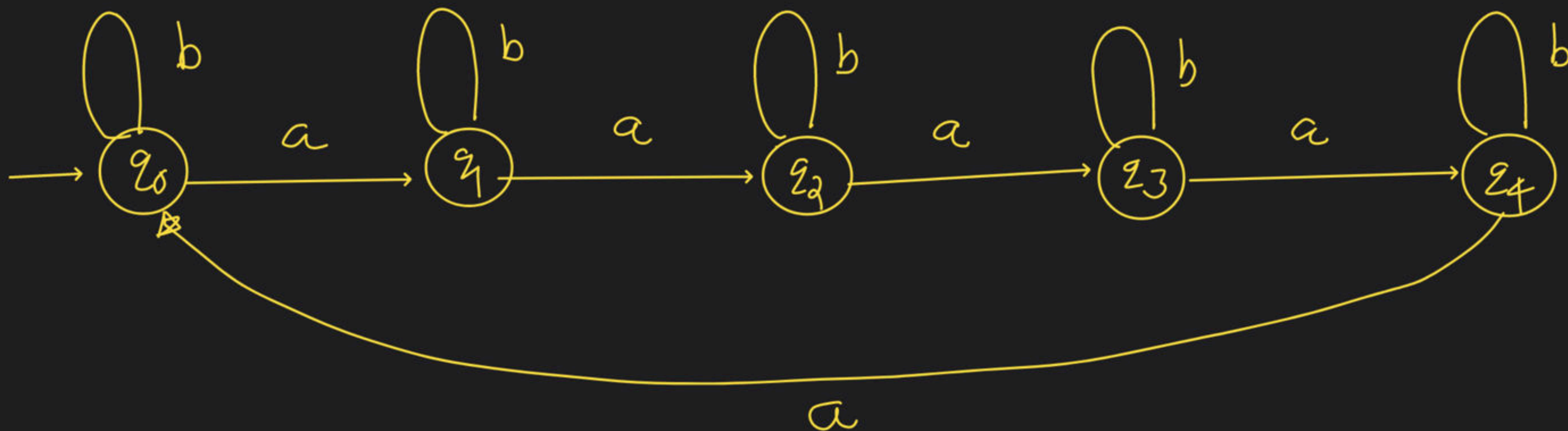
$$|w|_b \equiv 3 \pmod{4} = 3, 7, 11, 15, 19$$



$$|w|a \equiv 2 \pmod{5} = 2, 7, 12, 17,$$

Rem

0	1	2	3	4
q_0	q_1	q_2	q_3	q_4



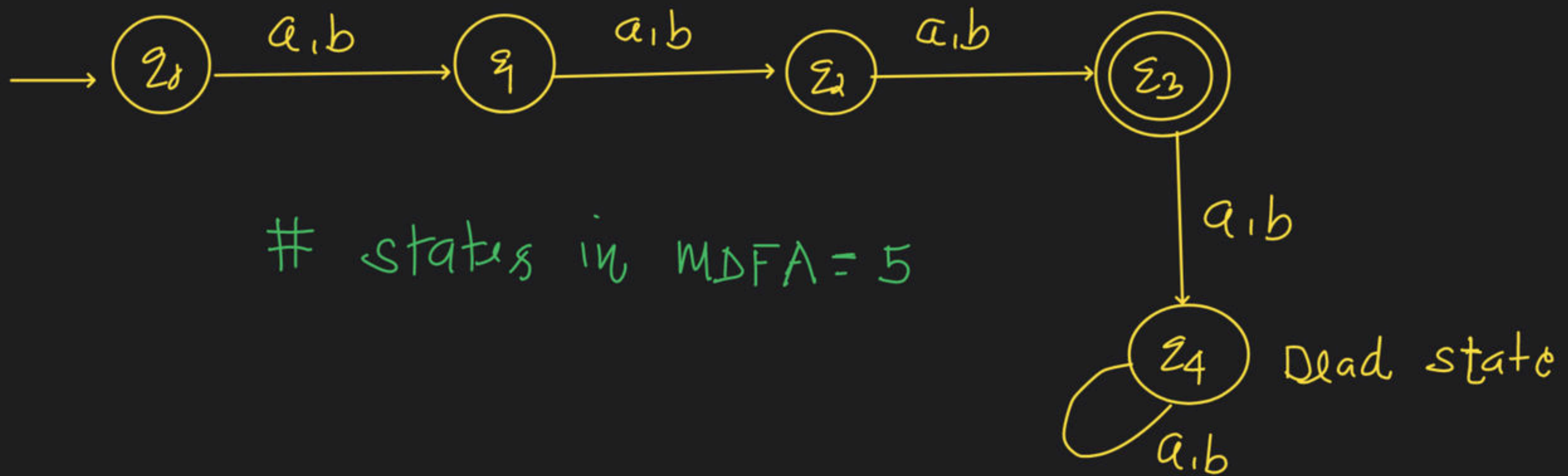
Let L is the set of all string of a's and b's where $|w|_a = r(\text{mod } n)$ or $|w|_b = r(\text{mod } n)$.
Here r represent the remainder and q_r represent the final state. number of
states in minimal DFA to accept this language is n.

Construct the minimal DFA that Accept all the string of a's and b's where length of the string

- i) Exactly 3 $\longrightarrow$ 5 state $\longrightarrow 3+2 = 5$ state
- ii) Atmost 3 $\longrightarrow$ 5 state $\longrightarrow 3+2 = 5$ state
- iii) Atleast 3 $\longrightarrow$ 4 state $\longrightarrow 3+1 = 4$ state

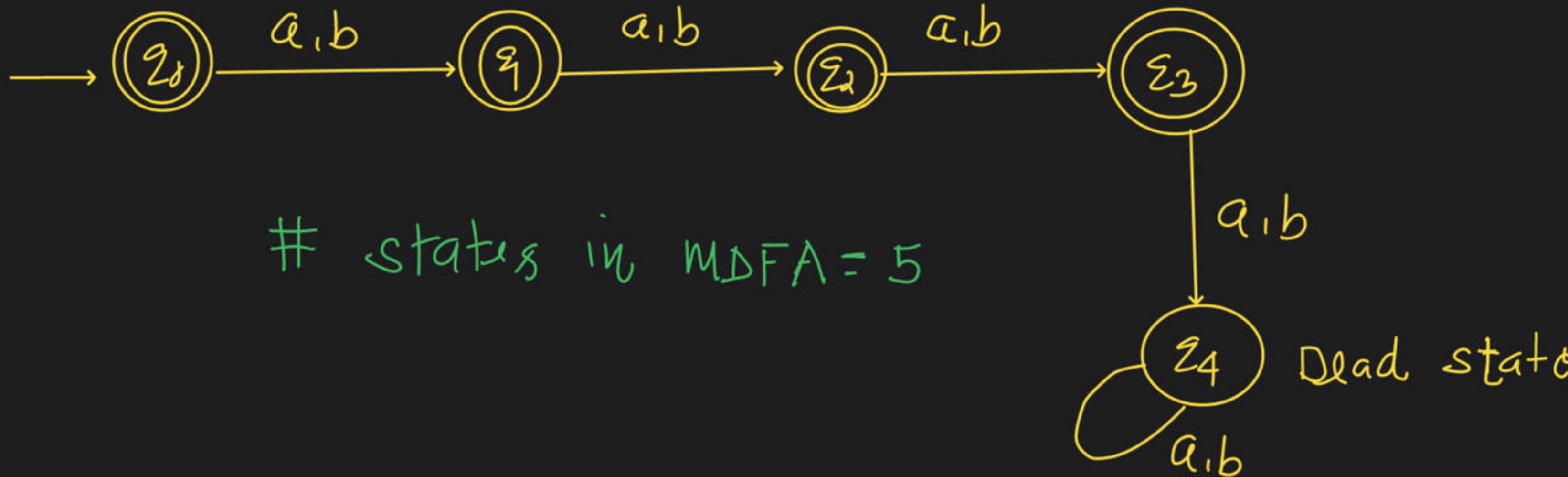
(i) unacademy $|w| = 3$

$|w| = x \ x \ x$
 ↓ ↓ ↓
 {a,b} {a,b} {a,b}



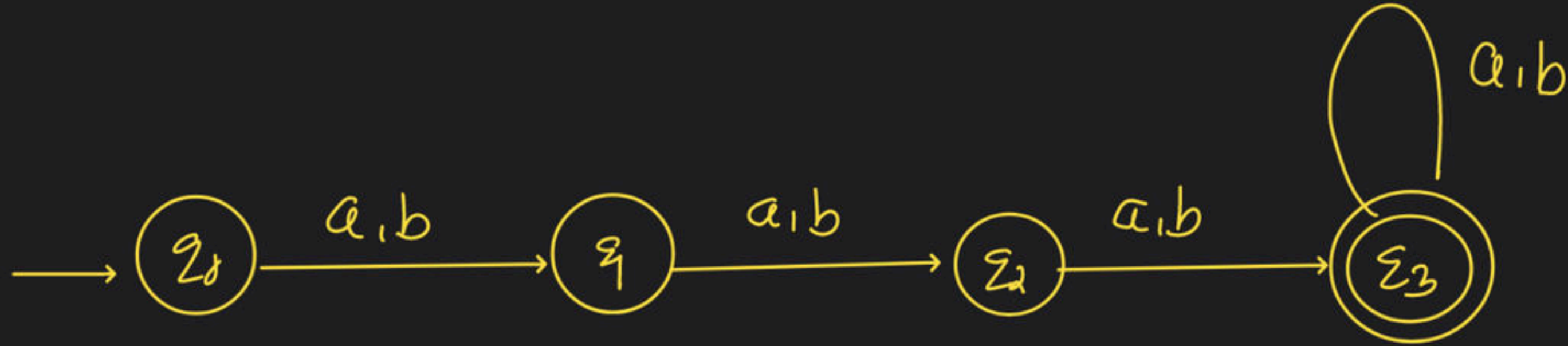
$$|w| \leq 3$$

↓ ↓ ↓ ↓
0 1 2 3



(iii) $|w| \geq 3$

$\downarrow \quad \downarrow \quad \downarrow$
3 4 5 -----



states in DFA = 4



Note {ADRule}

$$|w| = n \longrightarrow n+2 \text{ state}$$

$$|w| \leq n \longrightarrow n+2 \text{ state}$$

$$|w| \geq n \longrightarrow n+1 \text{ state}$$

Construct the minimum DFA that accept all the string of a's and b's where

- ✓ i) Length of string divisible by 3
- ✓ ii) $|w| \equiv 2 \pmod{4}$
- ✓ iii) $|w| \equiv 3 \pmod{5}$
- ✓ iv) $|w| \equiv 2 \pmod{6}$
- ✓ v) $|w| \equiv 4 \pmod{7}$

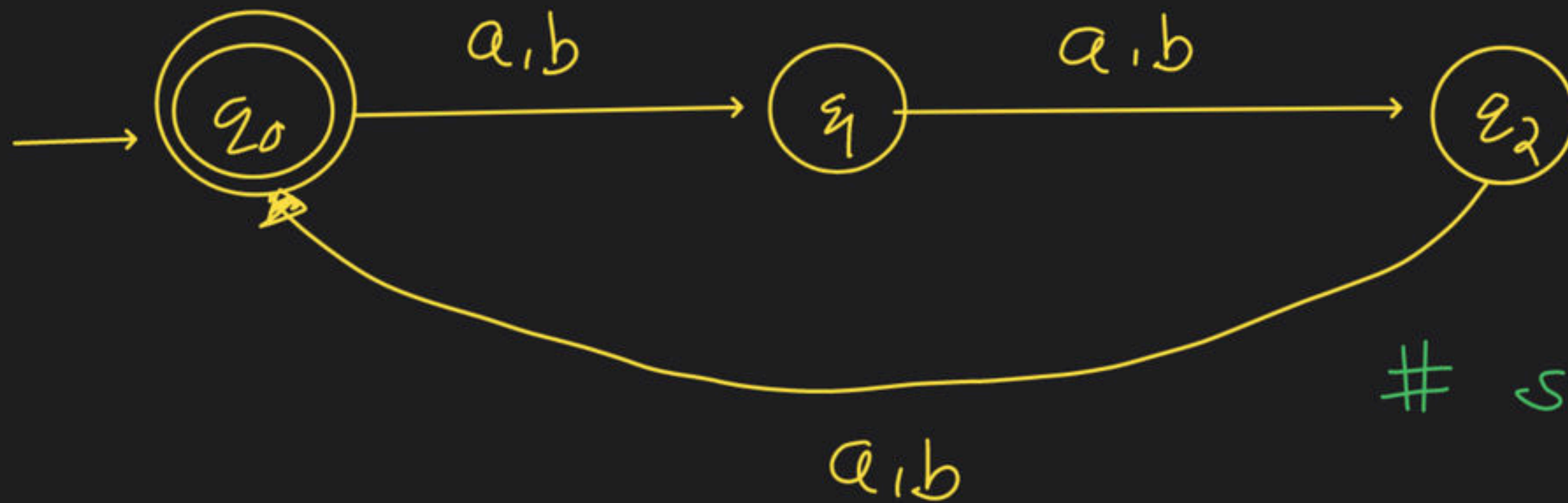


(1) soln

$$|w| = 0 \pmod{3} = 0, 3, 6, 9, 12$$

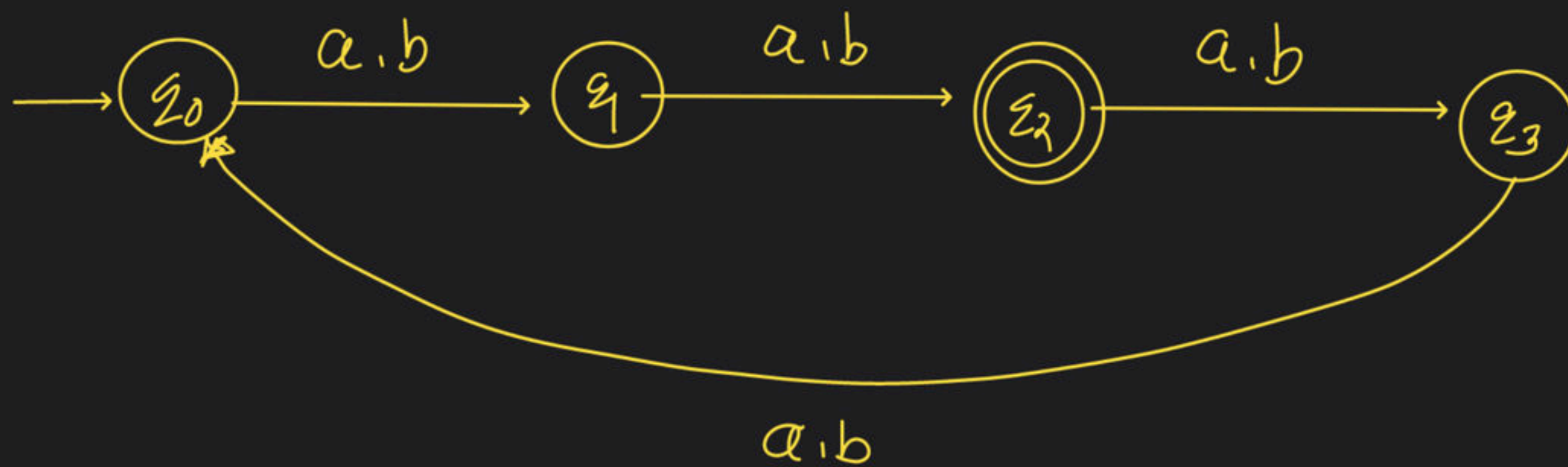
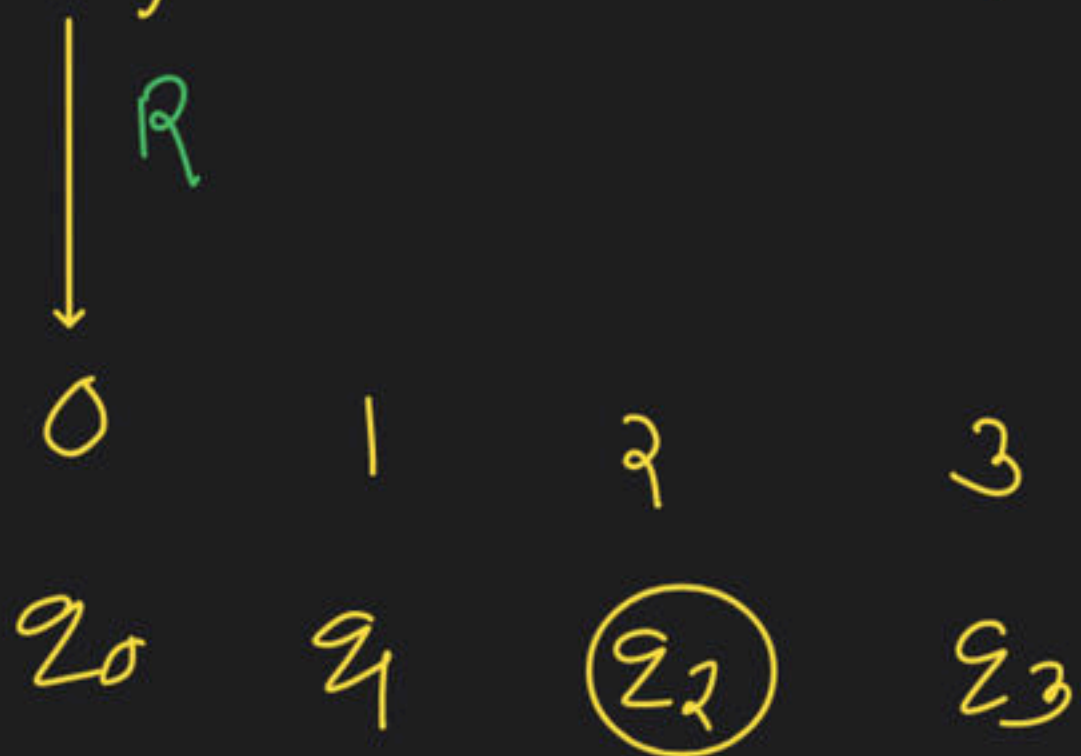
Rem

0	1	2
q_0	q_1	q_2

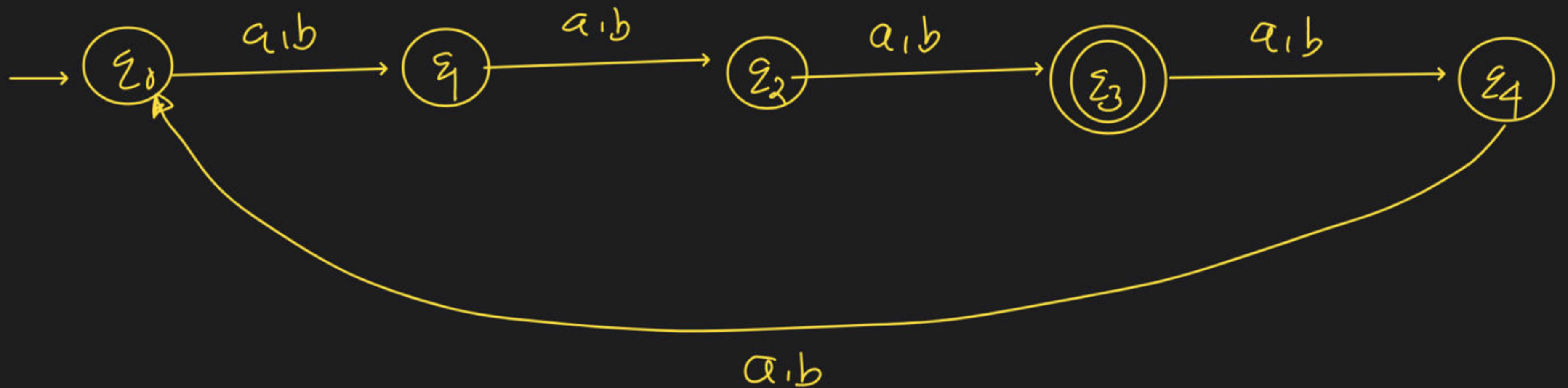
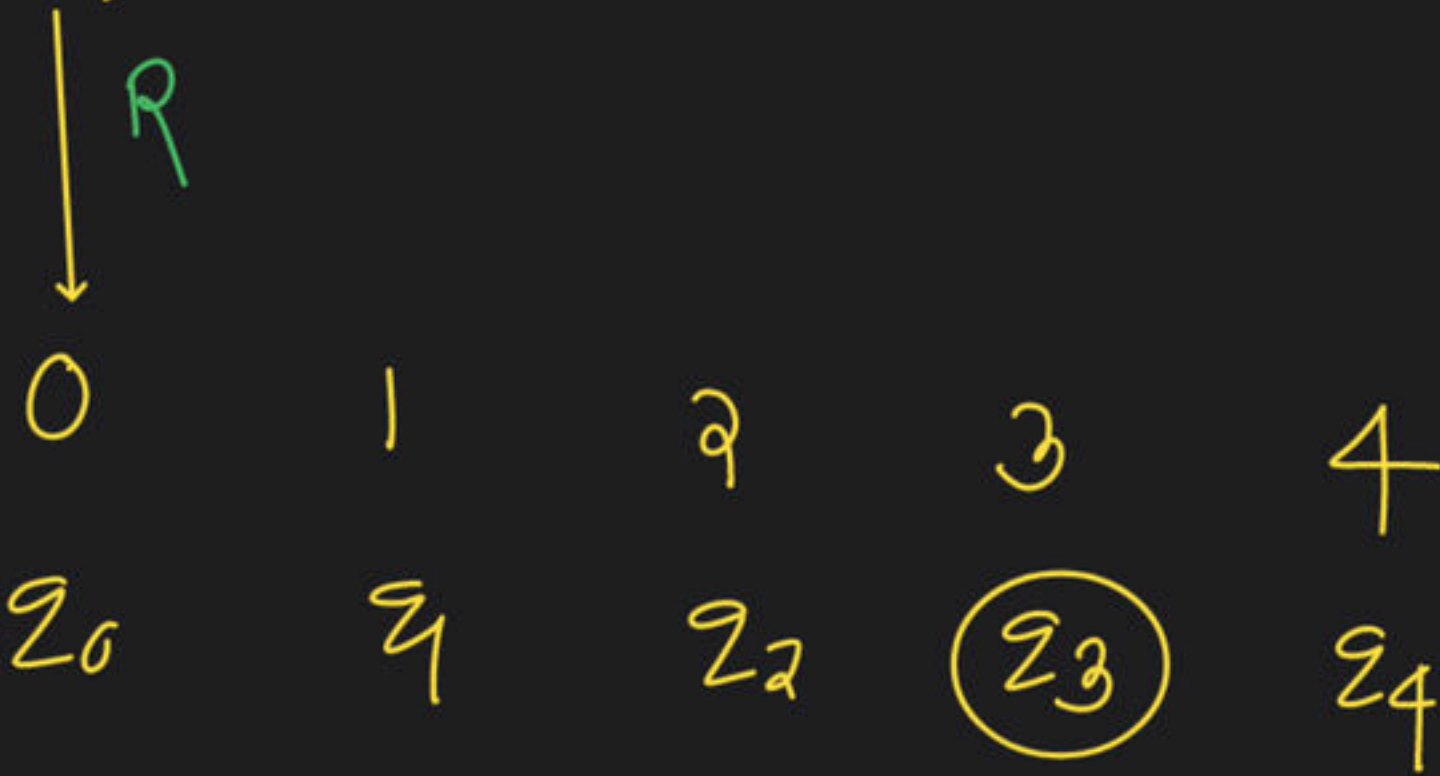


states in DFA = 3

$$|W| \equiv 2 \pmod{4} = 2, 6, 10, 14$$



(iii) $|w| \equiv 3 \pmod{5} = 3, 8, 13, 18$



Let L is the set of all string of a's and b's where $|w|=r(\text{mod } n)$. Here r represent the remainder and q_r represent the final state. number of states in minimal DFA to accept this language is n.



The smallest finite automaton which accepts the language $\{x \mid \text{length of } x \text{ is divisible by } 3\}$ has

$$|w| = 0 \pmod{3}$$



3 state

A

2 states

B

3 states

C

4 states

D

5 states



Thank You